

RISHABH KUMAR

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github.com/rishabh1244 | linkedin.com/in/rishabh-kumar-389334383 | Codeforces | LeetCode

TECHNICAL SKILLS

Languages: C, C++, Rust, JavaScript, TypeScript
Web & Backend: Next.js, React, Node.js, Express, MongoDB (MERN Stack)
Systems & Infra: Linux, Docker, Redis, Git, OpenGL, Raylib
Concepts: REST APIs, Containerization, Distributed Systems

EDUCATION

Indian Institute of Technology Madras 2025 – Present
BS Degree Program (Online) — Data Science & Applications GPA: **7.5/10**
BIT Sindri 2025 – Present
B.Tech in Civil Engineering

PROFESSIONAL EXPERIENCE

Freelance Software Developer 2025 – Present

- Designed and developed a production company website for a client, handling end-to-end delivery including UI design, frontend implementation, deployment, and post-launch maintenance.
- Delivered a performant, responsive web application using modern JavaScript frameworks while coordinating directly with stakeholders to meet design and functional requirements.

Open Source Contributor — Corsair.dev 2026

- Contributed to the open-source Corsair.dev platform by implementing an integration with Atlassian Confluence.
- Enabled users to connect and interact with Confluence-based documentation through the platform, extending its external service integrations.

KEY PROJECTS

Trading Engine — High-Performance Trading System | *Rust, Redis* GitHub

- Built a trading engine in Rust focused on low-latency order processing, market data handling, and concurrent execution.
- Designed modular components for order management, matching, and communication between trading services.
- Implemented performance-oriented data structures and concurrency patterns for efficient processing of high-throughput trading workloads.

Sketch.ai — Visual Learning Platform | *Next.js, LLM APIs, Excalidraw* sketch.ai

- Engineered a visual learning platform integrating Excalidraw with large language models to dynamically generate diagrams, animations, and flowcharts for physics, mathematics, and engineering concepts.
- Designed a prompt pipeline translating natural language queries into structured visual representations with step-by-step animation support.
- Architected a modular frontend enabling real-time collaborative whiteboard sessions with AI-generated overlays.

Break the Code — LLM Debugging Arena | *Next.js, Docker, LLM APIs*

- Developed an interactive platform where users intentionally introduce bugs into code while an LLM attempts to identify and repair them in real time.
- Containerized sandboxed code execution environments using Docker for secure, isolated, and scalable runtime evaluation of user-submitted code.
- Built a full-stack Next.js application managing LLM inference, session state, and sandboxed process lifecycle.

Sorting Algorithm Visualizer | *C, Raylib*

- Implemented a real-time algorithm visualization engine in C using Raylib, rendering step-by-step execution of merge sort, quicksort, and heapsort.
- Designed a frame-rate-controlled animation loop decoupling visual rendering from algorithmic state transitions to accurately demonstrate computational complexity.